Microsoft Defender HomeLab Extension - Notes

Date: 20 -21 October 2026
Status: Complete

Objective

Extend the existing hybrid HomeLab with Microsoft Defender for Business. Deploy endpoint security policies through Intune, investigate authorised test detections, practise remote response actions, and review security recommendations.

Preparation and onboarding

- Took a pre-Defender snapshot of CLIENT01.
- Completed Microsoft Defender for Business setup using Intune for policy management.
- Verified the Intune-Defender connection.
- Confirmed successful deployment of the EDR onboarding policy.
- Verified CLIENT01 appeared in Defender with an Active health state and an Onboarded status.
- Created the assigned security group LAB - Defender Devices and added CLIENT01.

Endpoint security policies

Configured and deployed the following policies:

Policy	Configuration
LAB - Defender Antivirus	Real-time, behaviour, cloud, script, download and archive scanning; safe sample submission; potentially unwanted application blocking
LAB - Defender Taper Protection	Enabled tamper protection; policy name currently contains this spelling
LAB - Defender Firewall	Enabled the Domain, Private and Public firewall profiles
LAB - ASR Audit	Set “Block process creations originating from PSExec and WMI commands” to Audit

Intune reported successful application of these policies. Windows Security showed antivirus and tamper protection enabled and managed by the administrator. Alice retained access to the Sales share after firewall deployment.

The ASR rule was configured in Audit mode; an actual ASR audit event has not yet been demonstrated.

EDR detection and incident investigation

Ran Microsoft's official Defender detection test on CLIENT01.

Defender generated incident #1, "Execution incident on one endpoint," containing a test PowerShell alert and a suspicious PowerShell command-line alert.

Reviewed:

- The incident attack story and affected device.
- The process tree and PowerShell command line.
- Evidence associated with the test, including the loopback address 127.0.0.1.
- The relationship between the observed activity and the published Microsoft test.

The reviewed evidence confirmed detection and reporting. It did not demonstrate blocking or quarantine.

Classified the incident as **Informational, expected activity — Security testing**, added an investigation comment, and resolved it.

EICAR antivirus test

Downloaded the harmless EICAR antivirus test file on CLIENT01.

Windows Security detected Virus:DOS/EICAR_Test_File in Alice's Downloads folder and reported **Status: Removed**.

Defender subsequently displayed incident #2, "'EICAR_Test_File' malware was prevented," with Informational severity.

This verified both local antivirus detection/removal and cloud reporting. Added a comment documenting the authorised test, classified it as security testing, and resolved the incident.

Remote antivirus scan

Submitted a remote Quick scan from CLIENT01's device page in Defender.

The portal initially showed the action as pending. Checked local scan timestamps using:

Get-MpComputerStatus | Select-Object QuickScanStartTime, QuickScanEndTime, FullScanStartTime, FullScanEndTime

The quick scan started at **22:01:22** and finished at **22:04:13**, lasting approximately **2 minutes 51 seconds**.

The portal subsequently reported **"Antivirus scan successfully triggered."** Local timestamps confirmed scan completion; the portal message confirmed successful triggering.

Device isolation and recovery

Used the Defender portal on the host PC to isolate CLIENT01.

- Left the optional Outlook, Teams and Skype communication exception unchecked.

- Confirmed Sales share access became blocked during isolation.
- Submitted a release-from-isolation action.
- Confirmed Alice could access Sales again afterward.
- Subsequently observed that the Isolated label had disappeared from CLIENT01’s header.

This demonstrated endpoint containment and restoration of normal access.

Vulnerability assessment and Windows patching

Defender initially reported **2,849 discovered vulnerabilities** and a High exposure level. Most visible findings affected Windows 11 build 26200.6584.

Windows Update had security update **KB5127648** pending restart. Restarted CLIENT01 and verified the new build using winver:

26200.6584 → 26200.9711

Afterward:

- Windows Update reported the device was up to date.
- Alice could still access the Sales share.

Local patch installation is verified. Defender reassessment confirmed vulnerabilities decreased from 2,849 to 57, a reduction of 2,792. The Windows 11 update recommendation disappeared. All remaining vulnerability findings relate to OpenSSL. CLIENT01 reports build 26200.9711, and Alice’s Sales access remains functional.

Windows patching is verified, OpenSSL open, Controlled folder access as Audit mode verified locally with portal reassessment pending.

OpenSSL recommendation review

Investigated “Update owning apps: vulnerable OpenSSL libraries detected.”

Defender listed 57 associated vulnerabilities and affected file paths within:

- Microsoft Office Hub, Photos and Paint.
- OneDrive installations in several user profiles.
- A Windows WebView component under WinSxS.

Completed the available Microsoft Store application updates.

Checked Alice’s OneDrive version: **26.183.0920.0005, 64-bit**. Microsoft’s release notes showed this as the Production version currently rolling out. No OneDrive reinstall or downgrade was performed.

The OpenSSL recommendation remains open. Store updates do not establish that all affected files, including copies in other user profiles, have been remediated.

Controlled folder access

Reviewed Defender’s recommendation to enable Controlled folder access or use Audit mode.

Updated the existing LAB - ASR Audit policy in Intune to configure **Controlled folder access in Audit mode**.

Audit mode records activity that would be blocked without enforcing those blocks.

Current status: Verified Audit mode on CLIENT01 using ‘Get-MpPreference | Select-Object EnableControlledFolderAccess’, which returned 2. The portal recommendation remained visible at the last check. Configuration is verified locally; an actual audit event has not been demonstrated.

Helpdesk

Ticket	Investigation and outcome
# 9	Confirmed the EICAR download disappeared because Defender detected and removed it.
# 10	Confirmed greyed-out antivirus controls were expected centrally managed settings. Protection remained enabled.
# 11	Removed a Linux inventory filter that hid CLIENT01. The device reappeared without onboarding changes.
# 12	A temporary outbound firewall rule blocked GLPI at 10.77.77.30:8080. Removing it restored the website and changed the TCP test from False to True.

All four tickets were assigned to Priya Shah, solved and accepted by Alice. Defender portal checks used the authorised lab administrator account. Priya’s GLPI assignment did not grant Defender permissions.

Outcome so far

CLIENT01 is onboarded to Microsoft Defender and receives centrally managed endpoint security policies. Authorised EDR and EICAR tests generated cloud incidents that were investigated and resolved. Remote antivirus scanning and device isolation/recovery were demonstrated successfully.

Windows patching updated CLIENT01 to build 26200.9711. Defender reassessment confirmed vulnerabilities decreased from 2,849 to 57, with all remaining findings relating to OpenSSL. Available Microsoft Store updates were installed, but OpenSSL remediation remains unverified.

Controlled folder access Audit mode was verified locally through PowerShell. All four Defender helpdesk exercises, recorded in GLPI tickets #9-#12, were resolved and accepted.

The practical work described above is complete. Final documentation and evidence review remain outstanding.

Extras

- Review the remaining OpenSSL findings and track application updates; these findings remain open.
- Recheck whether Defender’s Controlled folder access recommendation reflects the locally verified Audit mode.

Testing limitations

- ASR and Controlled folder access audit events have not been demonstrated.
- Resistance to an administrator’s attempted protection change has not been tested separately.
- Phishing/SmartScreen testing and automated investigation have not been demonstrated.